CHAPTER 4 Cell Biology

SYLLABUS

Prokaryotic and Eukaryotic Cell Structure; Cell Cycle and Cell Growth Control; Cell-Cell Communication, Cell Signaling and Signal Transduction.

CHAPTER ANALYSIS

Topic	GATE 2010	GATE 2011	GATE 2012	GATE 2013	GATE 2014	GATE 2015	GATE 2016	GATE 2017	GATE 2018	GATE 2019
Prokaryotic and Eukaryotic Cell Structure		3						1		1
Cell Cycle and Cell Growth Control		3				1	2	3	2	1
Cell-Cell Communication, Cell Signaling and Signal Transduction	2		3		2				1	

QUESTIONS

1. When thymidine synthesis is inhibited, eukaryotic cells will be arrested at

(A) end of S phase. **(B)** end of M phase.
(C) G_1/S interphase. **(D)** G_2/M interphase.

(GATE 2001; 1 Mark)

2. In the course of cell cycle, the level of the protein cyclin abruptly falls during

(A) G_1 phase. **(B)** S phase.
(C) G_2 phase. **(D)** M phase.

(GATE 2003; 1 Mark)

3. In the cell cycle of a typical eukaryote, the sequence of events operating at the time of cell division is

(A) S phase → G_2 phase → G_1 phase → M phase
(B) S phase → M phase → G_1 phase → G_2 phase
(C) S phase → G_2 phase → M phase → G_1 phase
(D) S phase → G_1 phase → M phase → G_2 phase

(GATE 2004; 2 Marks)

4. Match Group I with Group II.

Group I	Group II
P. Fibronectin	1. Uptake of amino acids and glucose
Q. Insulin	2. Trypsin inhibitor
R. α-Macroglobulin	3. Binds iron
S. Transferrin	4. Cell attachment to substratum

(A) P–2; Q–1; R–4; S–3
(B) P–3; Q–2; R–1; S–1
(C) P–4; Q–2; R–1; S–3
(D) P–4; Q–1; R–2; S–3

(GATE 2010; 2 Marks)

5. A mutant G_α protein with increased GTPase activity would

(A) not bind to GTP.
(B) not bind to GDP.
(C) show increased signaling.
(D) show decreased signaling.

(GATE 2010; 2 Marks)

6. Apoptosis is characterized by

(A) necrosis.
(B) programmed cell death.
(C) membrane leaky syndrome.
(D) cell cycle arrest process.

(GATE 2011; 1 Mark)

7. A cell in G_1 of interphase has 12 chromosomes. How many chromatids will be found per cell during metaphase II of meiosis?

(A) 6 (B) 12
(C) 18 (D) 24

(GATE 2011: 1 Mark)

8. The lipopolysaccharides present in bacterial cell wall has lipid A which is connected to

(A) O-polysaccharide.
(B) core polysaccharide.
(C) both with O-polysaccharide and core polysaccharide.
(D) rhamnose-mannose disaccharide.

(GATE 2011: 1 Mark)

9. Gas vacuoles are present in

(A) *Anabaena flos-aquae*.
(B) *Bacillus subtilis*.
(C) *Acanthurus nigrofuscus*.
(D) *Mycobacterium tuberculosis*.

(GATE 2011: 1 Mark)

10. Structured William's model

P. can describe the changes in intracellular components of the cell during growth.
Q. cannot describe the death phase of the cells.
R. can describe the variation of size of cells in the different phases of growth.
S. cannot describe the lag period of growth.

Which one of the following is correct?

(A) P, Q and S only
(B) P, Q and R only
(C) Q, R and S only
(D) P, R and S only

(GATE 2011: 2 Marks)

11. Determine the correctness or otherwise of the following Assertion (*A*) and the Reason (*R*).

Assertion: The elucidation of ribosome structure helps in the development of new generation drugs.

Reason: The high resolution of macromolecular structure has enabled in structure-based drug design.

(A) Both (*A*) and (*R*) are true and (*R*) is the correct reason for (*A*).
(B) Both (*A*) and (*R*) are true but (*R*) is not the correct reason for (*A*).
(C) (*A*) is true but (*R*) is false.
(D) (*A*) is false but (*R*) is true.

(GATE 2011: 2 Marks)

12. Choose the correct signal transduction pathway.

(A) Hormone → 7 TM receptor → G protein → [illegible] → PKA
(B) Hormone → G protein → 7 TM receptor → [illegible] → PKA
(C) Hormone → 7 TM receptor → G protein → [illegible] → cAMP
(D) Hormone → 7 TM receptor → cAMP → G protein → PKA

(GATE 2012: 1 Mark)

13. Match the entries in Group I with those in Group II.

Group I	Group II
P. MTT	1. Dihydrofolate reductase
Q. Annexin V	2. Succinate dehydrogenase
R. Methotrexate	3. Microtubules
S. Taxol	4. Phosphatidylserine

(A) P–3; Q–1; R–4; S–2
(B) P–2; Q–4; R–1; S–3
(C) P–2; Q–3; R–4; S–1
(D) P–4; Q–2; R–1; S–3

(GATE 2012: 2 Marks)

14. Match the products in Group I with the applications in Group II.

Group I	Group II
P. Digoxin	1. Muscle relaxant
Q. Stevioside	2. Anti-cancer agent
R. Atropine	3. Cardiovascular disorder
S. Vinblastine	4. Sweetener

(A) P–1; Q–4; R–3; S–2
(B) P–3; Q–2; R–1; S–4
(C) P–3; Q–4; R–1; S–2
(D) P–2; Q–3; R–1; S–4

(GATE 2012: 2 Marks)

15. Cholera toxin increases cAMP levels by

(A) modifying G_i protein.
(B) modifying G_s protein
(C) binding to adenylate cyclase.
(D) activating cAMP phosphodiesterase.

(GATE 2014: 2 Marks)

16. Match the items in Group I with Group II.

Group I	Group II
P. Receptor tyrosine kinase	1. Inactivation of G-proteins
Q. Cyclic GMP (cGMP)	2. Reception of insulin signal

Group I	Group II
R. GTPase activating protein (GAP)	3. Thyroid hormone
S. Nuclear receptor	4. Receptor guanylyl cyclase

(A) P–1; Q–3; R–4; S–2
(B) P–2; Q–4; R–3; S–1
(C) P–3; Q–1; R–4; S–2
(D) P–2; Q–4; R–1; S–3

(GATE 2014; 2 Marks)

17. The cytokinetic organelle in plant cells is

(A) centriole. (B) phragmoplast.
(C) proplastid. (D) chromoplastid.

(GATE 2015; 1 Mark)

18. In a typical mitotic cell division cycle in eukaryotes, M phase occurs immediately after the

(A) G_0 phase. (B) S phase.
(C) G_1 phase. (D) G_2 phase.

(GATE 2016; 1 Mark)

19. Which one of the following statements regarding G proteins is incorrect?

(A) GDP is bound to G protein in the resting stage.
(B) GTP bound α subunit cannot reassemble with $\beta\gamma$ dimer.
(C) All G proteins are trimeric.
(D) Activation of G protein may result in activation or inhibition of the target enzymes.

(GATE 2016; 2 Marks)

20. Secretory proteins synthesized by ER-associated ribosomes traverse through

(A) mitochondria. (B) peroxisomes.
(C) the Golgi apparatus (D) the nucleus.

(GATE 2017; 1 Mark)

21. In eukaryotes, cytokinesis is inhibited by

(A) cytochalasin D. (B) vinblastine.
(C) nocodazole. (D) colchicine.

(GATE 2017; 1 Mark)

22. Match the proteins in Group I with cellular processes in Group II.

Group I	Group II
P. p53	1. DNA packaging
Q. Lysozyme	2. Apoptosis
R. Tubulins	3. Hydrolysis of polysaccharides
S. Histones	4. Chromosome segregation

(A) P–4; Q–2; R–3; S–1 (B) P–2; Q–3; R–1; S–4
(C) P–4; Q–3; R–1; S–2 (D) P–2; Q–3; R–4; S–1

(GATE 2017; 2 Marks)

23. The genome is diploid at the end of which phases of a human mitotic cell cycle?

(A) G_2 and S (B) G_1 and M
(C) M and S (D) G_1 and G_2

(GATE 2017; 2 Marks)

24. If serum is removed from the growth medium of human embryonic kidney cell line (HEK), then the cells will

(A) proliferate faster.
(B) proliferate normally.
(C) undergo cell cycle arrest.
(D) undergo immediate apoptosis.

(GATE 2018; 1 Mark)

25. Cell type A secretes molecule X into the culture medium. Cell type B in the same culture responds to the molecule X by expressing protein Y. Which one of the following modes of signaling represents the interaction between A and B?

(A) Autocrine (B) Juxtacrine
(C) Paracrine (D) Intracrine

(GATE 2018; 1 Mark)

26. Which one of the following statements is true for actin?

(A) Actin filament is structurally polarized, and the two ends are not identical.
(B) *De novo* actin polymerization is a single-step process.
(C) The pointed end of the actin filaments is the fast-growing end.
(D) Actin forms spindle fibers during mitosis.

(GATE 2018; 1 Mark)

27. DNA synthesis in eukaryotes occurs during which phase of the mitotic cell cycle?

(A) M (B) G_1
(C) S (D) G_0

(GATE 2019; 1 Mark)

28. Match the organelles in Group I with their functions in Group II.

Group I	Group II
P. Lysosome	1. Digestion of foreign substances
Q. Smooth ER	2. Protein targeting
R. Golgi apparatus	3. Lipid synthesis
S. Nucleolus	4. Protein synthesis
	5. rRNA synthesis

(A) P–1; Q–3; R–2; S–5
(B) P–1; Q–4; R–5; S–3
(C) P–2; Q–5; R–3; S–4
(D) P–1; Q–3; R–4; S–5

(GATE 2019; 2 Marks)

ANSWER KEY

1. (C)	2. (D)	3. (C)	4. (D)	5. (D)	6. (B)	7. (B)	8. (B)	9. (A)	10. (D)
11. (A)	12. (A)	13. (B)	14. (C)	15. (B)	16. (D)	17. (B)	18. (D)	19. (C)	20. (C)
21. (A)	22. (D)	23. (B)	24. (C)	25. (C)	26. (A)	27. (C)	28. (C)		

ANSWERS WITH EXPLANATIONS

1. Topic: Cell Cycle and Cell Growth Control

Eukaryotic cells will be arrested in interphase between G_1/S, on inhibition of thymidine synthesis. Thymidine is used by cells in S phase for DNA replication. Thus, in its absence DNA synthesis will be halted and thus cells will be struck in the interphase.

Answer (C)

2. Topic: Cell Cycle and Cell Growth Control

The cyclin concentrations of each type of cyclin rise gradually and then fall abruptly over the course of the cell cycle. Gradual increase in cyclin concentration stems from increasing transcription of cyclin genes. Rapid fall in cyclin concentration is precipitated by a full-scale targeted destruction of the protein. The abrupt degradation of M and S cyclins part way through M phase depends on a large enzyme complex called Anaphase-Promoting Complex (APC). This tags the cyclins with a chain of ubiquitin which then degrades them via proteasome degradation. This also returns the Cdk linked to these cyclins to inactive state.

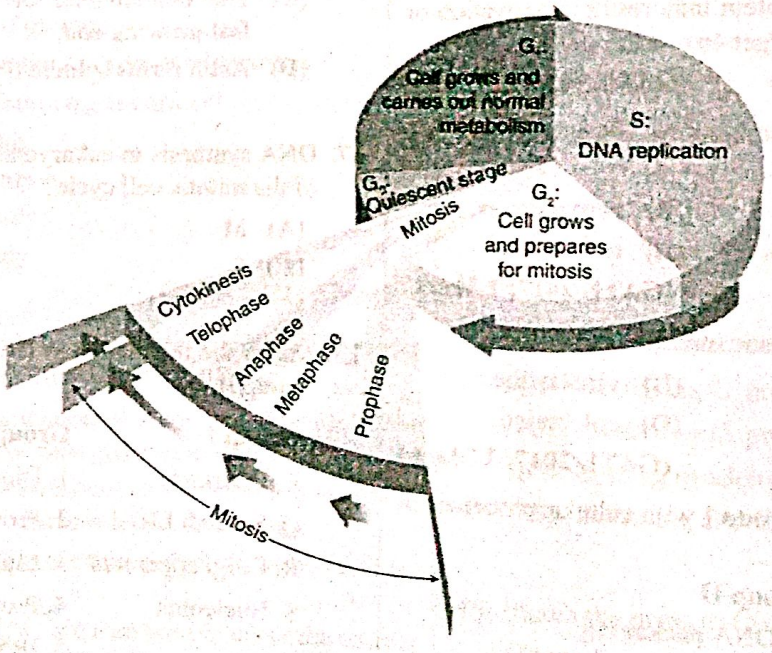
Answer (D)

3. Topic: Cell Cycle and Cell Growth Control

The eukaryotic cell cycle can be divided into following two major phases:

- Interphase: The interphase or the resting phase is the part of the cell cycle during which a cell grows, replicates cell organelles and DNA, assembles the "machinery" of mitosis and condenses its DNA. It is divided into Gap 1 (G_1) which is followed by Synthesis phase (S) and Gap 2 (G_2) phase.
- Mitosis (M) phase: In M phase, the duplicated chromosomes are separated into two nuclei (karyokinesis) and this is followed by cytokinesis.

Answer (C)



An overview of eukaryotic cell cycle.

4. Topic: Cell-Cell Communication

- Fibronectin facilitates a wide variety of cellular interactions with the extracellular matrix and plays significant roles in cell adhesion, migration, growth and differentiation processes including embryogenesis, wound healing, blood coagulation, host defense and metastasis.
- Glucagon, a peptide hormone, raises blood glucose level and insulin, a peptide hormone, lowers the blood glucose level. Insulin acts on various cells in

the body to accelerate facilitated transport of glucose into cells; and to increase the rate of conversion of glucose into glycogen (glycogenesis); to increase uptake of amino acids by cells and to increase protein synthesis; to speed synthesis of fatty acids (lipogenesis); to slow the conversion of glycogen to glucose (glycogenolysis); and to slow the formation of glucose from lactic acid and amino acids (gluconeogenesis) liver cells and adipose tissue. It also reduces the rate of catabolism of proteins and hence can be considered as an anabolic hormone.

- Alpha-2-Macroglobulin (α2M) is a large plasma protein which is found in the blood. It is largely produced by the liver, and is locally synthesized by macrophages, fibroblasts and adrenocortical cells. It is a pan-protease (including serine-, cysteine-, aspartic- and metalloproteinases) inhibitor involved in the clearance of proteins from the blood via endocytosis.
- Transferrin is a glycoprotein in the blood that plays an important part in binding to iron and transporting it between sites of absorption, storage and utilization in all vertebrate.

Answer (D)

5. Topic: Cell Signaling and Signal Transduction

A G protein is said to be "on" when its subunit is bound to GTP. G_α subunits can turn themselves off by hydrolysis of GTP to GDP and inorganic phosphate (P_i). This results in a conformational change causing a decrease in the affinity for the effector and an increase in the affinity for the subunit. Thus, following hydrolysis of GTP, the G_α subunit will dissociate from the effector and reassociate with the subunit to reform the inactive heterotrimeric G protein. Heterotrimeric G protein alpha subunits (G_α) is considered as regulatory GTPases. Thus, mutation in this heterotrimeric G proteins, leading to increased GTPase activity in G_α that turn G protein off and thus decrease signal transduction.

Answer (D)

6. Topic: Cell Cycle and Cell Growth Control

- Apoptosis refers to the programmed cell death that takes place in multicellular organisms. It is a highly regulated process characterized by specific morphological changes. These changes may include blebbing of cell membrane, cell shrinkage (i.e., condensation of nucleus and cytoplasm) followed by nuclear and cellular fragmentation.
- In contrast, necrosis is premature cell death often caused by external factors such as infections, trauma, toxin, etc. It is not a regulated pathway and doesn't follow any transduction pathway rather it results in loss of cell membrane integrity and uncontrolled release of cellular contents in extracellular space which ultimately leads to cell death.
- Cell cycle arrest process refers to the termination of DNA replication during S phase. It is usually caused by DNA damage and/or lack of nutrients. DNA damaged cells either undergo the process of DNA repair or apoptosis. Cells which lack nutrients may enter G_0 phase of cell cycle.

Answer (B)

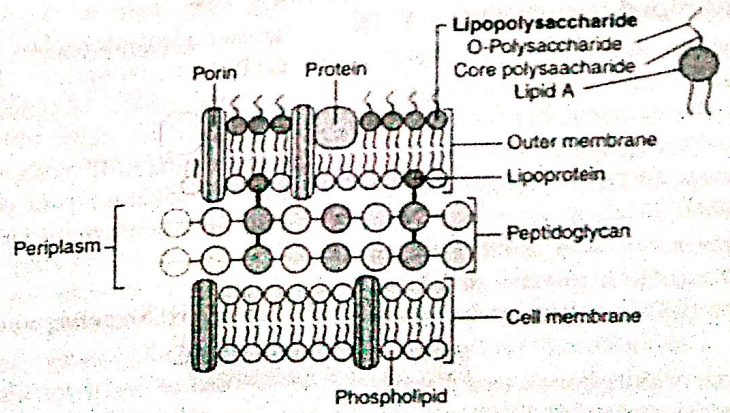
7. Topic: Cell Cycle and Cell Growth Control

A cell in G_1 of interphase has 12 chromosomes, i.e., $2n = 12$. As we know, meiosis I is a reductional division, hence, nucleus after first division must contain 6 chromosomes. In addition, meiosis II is same as mitosis and thus chromosome number remains same after or during metaphase II, i.e., 6 chromosomes or 12 chromatids.

Answer (B)

8. Topic: Prokaryotic Cell Structure

Lipopolysaccharides (LPS) are found in the outer membrane of Gram-negative bacteria. It consists of three parts: O polysaccharide or O antigen, core oligosaccharides and lipid A. Lipid A is covalently attached to the core oligosaccharide which is further connected to O antigen.



Gram negative bacteria cell wall

Answer (B)

9. Topic: Prokaryotic Cell Structure

Gas vesicles are rigid hollow structures which provide the function of buoyancy present in different microorganisms. For example, they are present in strains of *Anabaena flos-aquae*, *Gloeotrichia echinulata* and *Oscillatoria agardhii*.

Answer (A)

10. Topic: Cell Cycle and Cell Growth Control

William's model can be used to explain about the changes occurring in intracellular components of the cell during growth and variation in cell size during various phases of growth cycle. It cannot describe about the death phase.

Answer (D)

11. Topic: Eukaryotic Cell Structure

Information about the ribosome's structure is helping scientists to understand how many antibiotics attack certain parts, or functions, of the bacterial ribosome. Pharmaceutical and many biotechnology companies can avail this valuable knowledge to create new antibiotics to fight the growing issues of bacterial drug resistance. Structure-based drug design (SBDD, also known as rational drug design) is the technique that requires a high-resolution 3D structure of the inhibitor bound to the target obtained using X-ray crystallography.

Answer (A)

12. Topic: Cell Signaling and Signal Transduction

G protein-coupled receptors are known as 7 transmembrane (7-TM) domain receptors. Binding of neurotransmitter or primary messenger with 7-TM receptor activates G-protein which in turn activates adenylyl cyclase. It converts ATP to cyclic AMP, a secondary messenger, which helps in activating enzyme protein kinase A. PKA lastly brings out the biological effects such as gene expression, opening of ion channel, etc.

Answer (A)

13. Topic: Cell Signaling and Signal Transduction

- MTT [3-(4,5-Dimethylthiazol-2-yl)-2,5-Diphenyltetrazolium Bromide] is used to assess cell viability as a function of redox potential. MTT is reduced in the mitochondria of metabolically active cells by succinate dehydrogenase to yield a water-insoluble purple formazan crystal.
- Annexin V (or Annexin A5) is a member of the annexin family of intracellular proteins that binds to phosphatidylserine (PS) in a calcium-dependent manner.
- Methotrexate is a type of drug known as a disease-modifying anti-rheumatic drug (DMARD) and also a chemotherapeutic drug. For cancer, methotrexate competitively inhibits dihydrofolate reductase (DHFR), an enzyme that participates in the tetrahydrofolate synthesis. Dihydrofolate reductase (DHFR) catalyzes the reaction of 7,8-dihydrofolate and NADPH to form 5,6,7,8-tetrahydrofolate and $NADP^+$. Tetrahydrofolate is essential for the biosynthesis of purines, thymidylate, and several amino acids.
- Taxol (paclitaxel) is used to treat breast cancer, lung cancer, and ovarian cancer. Taxol blocks the cell cycle in its G_1 or M phases by stabilizing the microtubule cytoskeleton against depolymerization—the basis of its clinical use in cancer therapy.

Answer (B)

14. Topic: Cell Signaling and Signal Transduction

- Digoxin (Lanoxin), a drug used to treat congestive heart failure and the associated symptoms of shortness of breath.
- Stevioside is a natural sweetener extracted from leaves of *Stevia rebaudiana* (Bertoni).
- Atropine is an alkaloid of *Atropa belladonna* that competitively antagonizes the muscarinic action of acetylcholine and other cholinergic drugs.
- Vinblastine is a natural alkaloid obtained from the plant *Vinca rosea* Linn. It binds to tubulin and inhibits formation of microtubule, resulting in disruption of mitotic spindle assembly and arrest of tumor cells in the M phase of the cell division.

Answer (C)

15. Topic: Cell Signaling and Signal Transduction

Cholera toxin, an 87 kD protein of subunit composition AB5, binds to ganglioside GM1 on the surface of intestinal cells via its B subunits. This permits the toxin to enter the cell, probably via receptor-mediated endocytosis, where an ~195-residue proteolytic fragment of its A subunit is released. This fragment catalyzes the transfer of the ADP-ribose unit from NAD^+ to a specific Arg side chain of $G_{s\alpha}$. ADP-ribosylated $G_{s\alpha}$-GTP can activate adenylate cyclase but cannot hydrolyze its bound GTP. As a consequence, the adenylate cyclase is locked in its active state and cellular cAMP levels increase ~100-fold. Intestinal cells, which normally respond to small increases in cAMP by secreting digestive fluid (an HCO_3^- rich salt solution), pour out enormous quantities of this fluid in response to the elevated cAMP concentrations.

Answer (B)

16. Topic: Cell Signaling and Signal Transduction

- Insulin and many other polypeptide growth factors bind to receptors whose C-terminal domains have tyrosine kinase activity. Such receptor tyrosine kinases (RTKs) typically contain only a single transmembrane segment and are monomers or

loosely associated dimers in the unliganded state. When an RTK dimerizes or its conformation changes on ligand binding, its cytoplasmic protein tyrosine kinase (PTK) domains are brought close together so that they cross phosphorylate each other on specific Tyr residues. This autophosphorylation activates the PTK so that it can phosphorylate other protein substrates.

- Guanylyl cyclase is a family of enzymes that catalyze the conversion of GTP to cGMP. The family comprises both membrane-bound and soluble isoforms that are expressed in nearly all cell types. Stimulation of guanylyl cyclase and the resultant accumulation of cGMP regulates complex signaling cascades through immediate downstream effectors, including cGMP-dependent protein kinases, cGMP-regulated phosphodiesterases, and cyclic nucleotide-gated ion channels.
- GTPase activating protein (GAPs) are a family of regulatory proteins whose function is to accelerate the intrinsic rate of hydrolysis of bound GTP to GDP, which switches off GTPase signaling. They are characterized by a *rho* GAP domain, also known as the breakpoint cluster region-homology domain, of about 150 amino acids that fold into nine α-helices.
- Receptors for thyroid hormones are members of a large family of nuclear receptors that include those of the steroid hormones. They function as hormone-activated transcription factors and thereby act by modulating gene expression.

Answer (D)

17. Topic: Cell Cycle and Cell Growth Control

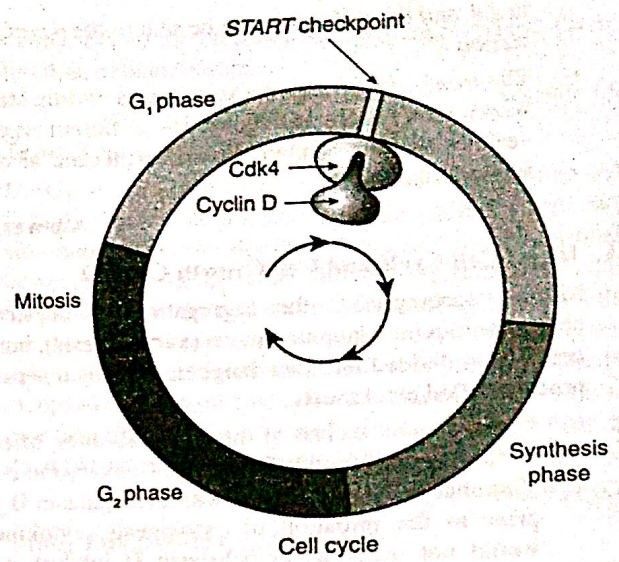
The plane in which the cell plate forms is though perpendicular to the axis of the mitotic spindle but is not determined by the position of the spindle unlike the case for animal cells. The spindle is retained for sometimes and is known as phragmoplast.

Answer (B)

18. Topic: Cell Cycle and Cell Growth Control

Cell cycle is divided into two main phases:

- M-phase includes mitosis (during which duplicated chromosomes are separated into two nuclei) and cytokinesis (entire cell divides into two daughter cells).
- Interphase is the period between cell divisions when cell grows, and several metabolic activities take place. G_2 (second gap) is a specific time period between end of DNA synthesis and beginning of M phase. S phase is the phase of cell cycle when replication occurs. G_1 (first gap) is the period following mitosis and preceding DNA synthesis.



Hence, as seen in the figure M phase occurs immediately after the G_2 phase.

Answer (D)

19. Topic: Cell Cycle and Cell Growth Control

G-proteins are heterotrimeric composed of G_α, G_β and G_γ subunits. Each subunit is encoded by a family of related genes. In its inactivated state, heteromeric G protein maintains its heteromeric state and its G_α subunit binds GDP. However, the binding of such G_α·GDP-$G_{\beta\gamma}$ complex to its cognate G-protein coupled receptor (GPCR) in complex with a ligand induces the G_α subunit to exchange its bound GDP for GTP.

When GTP is bound to G_α, its γ phosphoryl group promotes conformational changes in three of G_α's so-called switch regions, causing G_α to dissociate from $G_{\beta\gamma}$. This occurs because the binding of GTP's γ phosphoryl group and the binding of $G_{\beta\gamma}$ to G_α are mutually exclusive.

Answer (C)

20. Topic: Eukaryotic Cell Structure

The various proteins required for building of the membranes of the different organelles or those secreted outside the cells, are sorted and transported through the endoplasmic reticulum (ER) and Golgi apparatus, in both plant and animal cells. In plant cells, the Golgi apparatus also serves as the site for synthesis of the polysaccharides required for making the cell wall. In Golgi complex consists of several compartments. A series of large, flattened cisternae are present that are divided into *cis*, *medial* and *trans* cisternae.

- The *cis*—most face is closed to the ER. It severs as a sorting station for proteins that need to be sent back

to ER and those that need to be sent to the next Golgi station.

- The *trans*—most region also acts as a sorting station where proteins are segregated into different types of vesicles that are sent to plasma membrane or other intracellular parts.

Answer (C)

21. Topic: Cell Cycle and Cell Growth Control

Mitosis accomplishes the segregation of duplicated chromosomes into daughter nuclei (karyokinesis), but the cell is also divided into two daughter cells by a separate process called cytokinesis.

- Cytochalasins, a class of fungal metabolites, affect a wide variety of motile functions of eukaryotic cells. If animal cells were treated with cytochalasin B just prior to the initiation of cytokinesis, cytokinesis would not occur as cytochalasin B inhibits actin microfilaments. If plant cells are treated with cytochalasin B just prior to cytokinesis, it will occur normally since cytokinesis in plants does not appear to depend on actin filaments. Cytochalasin D inhibits protein synthesis.
- Vinblastine prevents cell division during anaphase of mitosis by binding to the microtubular proteins.
- Nocodazole arrests mitosis at G_2 or M phase.
- Colchicine affects cell division by its disruptive action on the mitotic spindle.

Answer (A)

22. Topic: Cell Cycle and Cell Growth Control

- p53, also known as TP53 gene encodes proteins that restrain cell growth and prevent cells from becoming malignant. In human cancer, the product of tumor suppressor gene may be able to suppress cancer formation by several different mechanisms.
- Lysozyme is the enzyme that is present in tears, saliva and sweat. It kills bacteria by breaking the chemical bonds in the outer cell wall of the bacteria, cleaves the covalent linkage in sugars present in peptidoglycan, and specifically targets Gram-positive bacteria.
- Tubulin, the monomers of microtubules are globular proteins arranged in longitudinal rows, termed protofilaments. Tubulin subunits are added to the plus ends of microtubules during metaphase, keeping the length of the chromosomal fibers constant, whereas subunits are lost from the plus ends during anaphase, resulting in shortening of the chromosomal fibers. Thus, they play a key role in chromosome segregation.
- Histones play structural role in chromatin. Their amount in chromatin is equivalent to that of DNA which indicates that the interaction between histones and DNA is conserved in eukaryotes. The conserved structure indicates that histones have an important role in packaging of DNA.

Answer (D)

23. Topic: Cell Cycle and Cell Growth Control

The cell cycle can be divided into two major phases based on cellular activities:

A. Interphase: The interphase is divided into the following three phases:

- Gap 1 (G_1) phase: It represents the time gap between the last cell division and the start of DNA replication. It is also known as first growth phase, post-mitotic phase and pre-synthetic phase. During this stage of interphase, the cell doubles its size and carries out its normal life. In this phase, the cell is growing and performs metabolically active functions. It is also characterized by formation of biochemical and cellular organelles like mitochondria, Golgi apparatus, ribosomes, etc.
- Synthesis (S) phase: During this stage, the DNA of the cell is replicated in the nucleus in preparation for cell division and its amount doubles. The chromosome number, however, remains the same. The cell synthesizes the additional histones that will be needed as the cell doubles the number of nucleosomes (repeating subunits of DNA and histones) in its chromosomes. The centriole present in the cytoplasm also duplicates. By the end of S phase chromosome changes from monad (single-stranded) form to dyad (double-stranded) form.
- Gap 2 (G_2) phase: It represents the time gap between the end of DNA replication and the beginning of cell division. It is also known as second growth phase, pre-mitotic phase and post-synthetic phase. During this phase, the supercoils of DNA condense into tightly compacted bodies that become visible as chromosomes during mitosis. Protein synthesis also takes place in the G_2 phase which is required for preparation of mitosis. It is also accompanied by cell growth.

B. Mitosis (M) phase: In M phase, the duplicated chromosomes are separated into two nuclei and this is followed by cytokinesis during which the entire cell divides into two daughter cells. These are diploid and contain identical sets of chromosomes and exactly the same information as parent cell.

Answer (B)

24. Topic: Cell Cycle and Cell Growth Control

Cells grow, divide, multiply and proliferate in the growth medium. For this, cells require certain growth factors such as mitogen growth factor (MGF). Serum contains these growth factors and certain hormones that direct

nucleus to produce genes required for the growth and division of the cells. Thus, removing of serum from medium would interrupt cell cycle and stop its growth and proliferation.

An appropriate growth or culture medium is required to support the growth of cells. In animal cell culture, serum is one of the most common supplements. It has growth factors, hormones and growth inhibitors. It also provides amino acids, vitamins, carbohydrates, lipids and minerals. Serum also acts as a buffer in the medium.

Answer (C)

25. Topic: Cell-Cell Communication

In the given example, since cell type A releases X that acts on neighboring cells B in the medium that are not in close context with A, paracrine signaling is taking place. Cells communicate with each other through extracellular messenger molecules in autocrine signaling, the molecules act on the same cell that secreted them. In juxtacrine signaling, close contact of cells is required. In paracrine signaling molecules act on neighboring cells. In intracrine signaling the molecule acts within the cell synthesizing it.

Answer (C)

26. Topic: Cell Cycle and Cell Growth Control

Actin filaments or microfilaments are solid, thinner structures, often organized into a branching network and composed of globular subunit of the protein actin. They are contractile and are associated with protein myosin for contraction. Each actin subunit also possesses polarity, that is, the two ends of an actin filament have different structures and dynamic properties. This results in a defined polarity of the microfilaments as well. Actin monomers, in the presence of ATP, polymerize to form a flexible, helical filament.

Answer (A)

27. Topic: Cell Cycle and Cell Growth Control

During the S stage of interphase, the DNA of the cell is replicated in the nucleus in preparation for cell division and its amount doubles. The chromosome number, however, remains the same. The cell synthesizes the additional histones that will be needed as the cell doubles the number of nucleosomes (repeating subunits of DNA and histones) in its chromosomes. The centriole present in the cytoplasm also duplicates. By the end of S phase chromosome changes from monad (single-stranded) form to dyad (double-stranded) form.

Answer (C)

28. Topic: Eukaryotic Cell Structure

- Lysosomes are vesicular structures containing several digestive enzymes and help in cell's waste disposal, by digesting compounds that enter the cell by phagocytosis or endocytosis.
- The smooth endoplasmic reticulum is the main site for the synthesis of fatty acids, steroids, and lipids.
- An important function of the Golgi apparatus is to sort the wide variety of macromolecules produced by the cell and target them for distribution to their proper location. Specialized molecular identification labels or tags, such as phosphate groups, are added by the Golgi enzymes to aid in this sorting effort.
- Nucleolus contains RNA, enzymes and proteins called histones. They synthesize ribosomal RNA, assemble ribosomal subunits.

Answer (C)